1 Experimental Setup

1.1 Design principles

The numerical experiment is designed to answer a single question:

> Does enabling the quasi-Newton fallback improve the reliability of thimble-HMC runs under a fixed computational budget?

To isolate this effect, all numerical comparisons are made under the following principles:

- only one algorithmic factor is changed at a time,
- paired random seeds are used whenever possible,
- the compute budget is fixed in advance,
- pass/fail criteria are specified before inspecting the results.

1.2 Two-level design

We use a two-level design consisting of

- a primary A/B comparison at a single production point,
- a small robustness sweep around that point.

The primary A/B comparison is the main result. The sweep is included only to test whether the main conclusion is stable under modest parameter variation.

1.2.1 Primary A/B comparison

The only algorithmic factor changed is

$$\text{quasi-fallback} \in \{\texttt{ON}, \texttt{OFF}\}.$$

All other settings are fixed at the production point:

$$\texttt{trajectory\_length} = 2, \qquad \texttt{integration\_steps} = 20, \qquad \texttt{initial\_flow\_time} = 0.4.$$

The two arms of the comparison use

- the same number of chains,
- identical stopping rules,
- identical solver tolerances,
- paired random seeds (common random numbers).

The run length is determined by a fixed compute budget rather than by an adaptive stopping criterion. This avoids bias caused by early stopping or selective truncation.

The primary comparison uses

$$10 \text{ paired-seed replicates.}$$

1.2.2 Robustness sweep

To test whether the primary conclusion depends strongly on the production-point choice, we perform two one-factor sweeps. Each point uses

3 paired-seed replicates.

Sweep A: integration steps

$$\texttt{integration\_steps} \in \{10, 20, 40\}, \qquad \texttt{trajectory\_length} = 2, \qquad \texttt{initial\_flow\_time} = 0.4.$$

Sweep B: initial flow time

$$\texttt{initial\_flow\_time} \in \{0.2, 0.4, 0.6\}, \qquad \texttt{trajectory\_length} = 2, \qquad \texttt{integration\_steps} = 20.$$

These sweeps are not intended as optimization studies. Their role is only to assess qualitative robustness.

1.3 Primary endpoint

The primary endpoint of the experiment is the *pass rate* across paired seeds.

A run is counted as pass if it satisfies all pre-specified convergence and physics-consistency conditions listed below. The main comparison is therefore

$$\text{pass rate (fallback ON)} \quad \text{vs} \quad \text{pass rate (fallback OFF)}.$$

This endpoint is chosen because a nominally faster run is not useful if it fails to produce a trustworthy chain.

1.4 Secondary diagnostics

In addition to the primary endpoint, we record secondary diagnostics for interpretation.

1.4.1 Convergence diagnostics

$$\hat{R}_{\max} = \max_{k \in \mathcal{C}} \hat{R}_k,$$

$$\mathrm{ESS}^{\mathrm{bulk}}_{\min} = \min_{k \in \mathcal{C}} \mathrm{ESS}^{\mathrm{bulk}}_k,$$

$$\mathrm{ESS}^{\mathrm{tail}}_{\min} = \min_{k \in \mathcal{C}} \mathrm{ESS}^{\mathrm{tail}}_k.$$

Here $\hat{R}_k$ denotes the split, rank-normalized $\hat{R}$ for component $k \in \mathcal{C}$.

1.4.2 Physics-consistency checks

We monitor the virial identity and the exact expectation value of z.

$$Z_{\mathrm{vir}} = \frac{|\langle \mathrm{virial} \rangle|}{\sigma(\langle \mathrm{virial} \rangle)}, \qquad Z_z = \frac{|\langle z \rangle - z_{\mathrm{exact}}|}{\sigma(\langle z \rangle)}.$$

These are dimensionless discrepancy measures in units of the estimated standard error.

1.4.3 Solver behavior

For each run we record

- Newton-only success rate,
- fallback invocation rate,
- hard-fail rate.

1.4.4 Sampling efficiency

We report

$$\mathrm{ESS/sec}$$

for the main observables.

Efficiency is treated as a secondary quantity because it is meaningful only after reliability has been established.

1.5 Pass/fail criteria

A run is counted as pass if and only if all of the following conditions hold:

$$\hat{R}_{\max} \leq 1.01,$$

$$\mathrm{ESS}^{\mathrm{tail}}_{\min} \geq 400,$$

$$Z_{\mathrm{vir}} \leq 2, \qquad Z_z \leq 2.$$

Runs failing any of these conditions are classified as failures.

1.6 Statistical summaries

The primary comparison will be summarized by

- pass rate across paired seeds,
- median $\hat{R}_{\max}$,
- median $\mathrm{ESS}^{\mathrm{tail}}_{\min}$,
- median hard-fail rate.

Where useful, paired differences

$$(\mathrm{ON} - \mathrm{OFF})$$

will also be reported for seed-matched runs.

1.7 Tables and figures

The numerical results will be presented using the following pre-specified summaries.

- **Main table:** pass rate across seeds (ON vs OFF), together with median $\hat{R}_{\max}$, $\mathrm{ESS}^{\mathrm{tail}}_{\min}$, and hard-fail rate.
- **Scatter plot:**

$$\hat{R}_{\max} \text{ vs } Z_{\mathrm{vir}}.$$

- **Sweep heatmap:** pass-rate difference

$$(\mathrm{ON} - \mathrm{OFF})$$

  for the robustness sweep.
- **Optional diagnostic panel:** running estimates of $\hat{R}$ and $\langle\text{virial}\rangle$ for one representative paired seed.